

SecureGate – Notebook 03: Logical H_L and S_L Gate Fingerprints

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Purpose. This notebook extends the X_L/Z_L baseline from Notebook 02 to fault-tolerant logical H_L and S_L operations. The objective is to determine whether the installed **surface-sim** implementation produces gate-dependent structure in the spacetime detector record, and whether that structure can support gate inference. The notebook proceeds from backend inspection, through encoded-circuit verification and detector-fingerprint analysis, to a Bernoulli Naive Bayes inference baseline with statistical and fresh-data validation.

1. Experimental configuration. The experiments use the unrotated CSS surface-code mapping because the installed backend exposes both logical H and S through `set_fold_trans.h()` and `set_fold_trans.s()`. A distance-5 patch contains

$$41 \text{ data qubits} + 40 \text{ ancilla qubits} = 81 \text{ physical qubits},$$

encoding one logical qubit, $L0$. The physical error probability is

$$p = 10^{-3},$$

and detector records are generated with a pre-gate detector frame.

The gate helpers modify the layout in place rather than directly executing the logical operation. Using the common physical corner data qubit $D1 = (0, 0)$ as the folding reference, the installed implementation stores reflection-dependent physical operations and transformed stabilizer information. For H_L , the implementation uses local physical H operations together with reflection-dependent SWAP pairings. For S_L , it uses local $S/S^\dagger$ operations together with reflection-dependent CZ connections. These are properties of the specific **surface-sim** construction used here, not universal requirements for every surface-code implementation.

Independent layout copies were used for H_L and S_L so that the in-place configuration of one experiment could not modify the other.

2. Logical circuits and encoded experiments. The high-level circuits are identical in structure:

$$|0\rangle_L \longrightarrow G_L \longrightarrow \text{measurement}, \quad G_L \in \{H_L, S_L\}.$$

They are represented as initialization (R), a logical gate separated by **TICK**, a second **TICK**, and final logical measurement (M). The configured layouts, independent noise models, detector objects, and the unrotated gate mapping are then passed to `experiment_from_circuit()`.

The encoded experiments contain the following verified resources:

	H_L	S_L
Physical qubits	81	81
Measurements	121	121
Detectors	100	100
Logical observables	1	1
TICKs	28	27

The detector-coordinate dictionaries match exactly between the two experiments. Both therefore use the same detector indexing and spacetime frame, with

$$t \in \{0.0, 1.0, 1.5\},$$

containing 40, 40, and 20 detectors, respectively. This establishes a one-to-one detector correspondence before any statistical comparison.

3. Detector sampling and fingerprint construction. Each encoded experiment was sampled for 10,000 shots, producing two binary detector matrices of shape

$$(10,000, 100).$$

For detector j , the empirical event probability is

$$\hat{P}(D_j = 1 \mid G) = \frac{1}{N} \sum_{k=1}^N D_j^{(k)}, \quad N = 10,000.$$

The detector fingerprint is therefore the resulting 100-dimensional vector of marginal detector-event probabilities. Both gates produced nonzero empirical event rates at 80 of the 100 detector locations. The mean event probabilities were

$$\overline{P}_{H_L} = 0.024598, \quad \overline{P}_{S_L} = 0.026675.$$

The detector-wise difference was defined as

$$\Delta_j = \left| \hat{P}(D_j = 1 \mid H_L) - \hat{P}(D_j = 1 \mid S_L) \right|.$$

Across all 100 detectors,

$$\overline{\Delta} = 0.003681, \quad \max_j \Delta_j = 0.019200,$$

with the maximum at detector 52, located at $(x, y, t) = (6, 1, 1.0)$.

The temporal decomposition shows strong localization of the difference:

t	mean $ \Delta $	max $ \Delta $
0.0	0.001070	0.006000
1.0	0.007425	0.019200
1.5	0.001415	0.004200

Thus, the largest marginal fingerprint differences occur at $t = 1.0$. The complete 100-dimensional profiles have Pearson correlation $r = 0.950529$, Euclidean distance 0.061401, and L_1 distance 0.368100. The high correlation indicates similar overall profiles, while the nonzero distances show measurable detector-level differences.

4. Gate inference from complete detector records. To test whether these differences are usable for inference, the complete detector records from both gates were combined into a balanced dataset:

$$20,000 \text{ records} = 10,000 H_L + 10,000 S_L.$$

A stratified 80/20 split produced 16,000 training and 4,000 test records. A Bernoulli Naive Bayes classifier was used as a simple baseline on the 100 binary detector features. This model assumes conditional independence of detector features given the gate class; therefore it is a baseline rather than an optimal inference method.

For the seed-42 test split, the accuracy was

$$A = 0.5527,$$

with confusion matrix

$$\begin{pmatrix} 1290 & 710 \\ 1079 & 921 \end{pmatrix}.$$

The corresponding class-specific recalls were 0.6450 for H_L and 0.4605 for S_L . The two-sided 95% Wilson confidence interval for the accuracy was

$$[0.5373, 0.5681],$$

and the exact one-sided binomial test against the balanced-class chance level $A_0 = 0.5$ gave

$$p = 1.3449 \times 10^{-11}.$$

The result is therefore statistically above chance for this test set, but the absolute gain is only 5.27 percentage points and is asymmetric between the two classes.

5. Robustness and controls. The notebook does not rely on the single seed-42 split. A repeated 10-split evaluation using stratified 80/20 partitions produced a mean accuracy of

$$\overline{A} = 0.5362, \quad \sigma_A = 0.0055,$$

with a range of 0.5270–0.5430. A separate five-fold cross-validation check gave a mean accuracy of 0.5340 with standard deviation 0.0085.

An independent fresh-shot validation was also generated directly from the encoded H_L and S_L experiments. The frozen classifier was evaluated on 10,000 new records (5,000 per gate), obtaining

$$A_{\text{fresh}} = 0.5374,$$

with 95% Wilson interval

$$[0.5276, 0.5472],$$

and one-sided exact binomial $p = 3.9069 \times 10^{-14}$. Across five repeated fresh-validation rounds, the accuracies were

$$0.5270, 0.5487, 0.5443, 0.5465, 0.5425,$$

with mean 0.5418 and standard deviation 0.0086.

As a label-permutation control, the detector records were kept unchanged while the gate labels were randomly permuted. The resulting accuracy was 0.5002, approximately the two-class chance level. This control supports the interpretation that the original above-chance performance is associated with the actual gate-record correspondence rather than the classification procedure alone.

6. Where the inference signal resides. The classifier was also evaluated using detectors from each temporal layer independently. Across 10 repeated splits, mean accuracies were

t	mean accuracy	standard deviation
0.0	0.5036	0.0060
1.0	0.5351	0.0059
1.5	0.5015	0.0058

The $t = 1.0$ layer therefore carries the strongest isolated classification signal and agrees with the earlier fingerprint-difference analysis. However, a paired comparison between the complete 100-detector record and the $t = 1.0$ subset gave mean accuracies of 0.5355 and 0.5370, respectively, with paired $t = -0.894$ and $p = 0.395$. Thus, restricting the model to $t = 1.0$ does not produce a statistically detectable change in accuracy across these paired splits.

The learned Bernoulli model is also consistent with the direct fingerprint analysis: the correlation between empirical and learned signed detector differences is 0.988210, with mean absolute profile errors of 0.000506 for H_L and 0.000524 for S_L . The largest learned detector contrasts occur at $t = 1.0$, led by detectors 47 and 52.

7. Scientific interpretation and limitations. The notebook establishes a complete implementation-to-inference chain: (i) inspect the installed logical-gate backend, (ii) construct independent fault-tolerant H_L and S_L experiments, (iii) verify a common detector spacetime frame, (iv) measure gate-dependent detector fingerprints, and (v) test whether complete detector records contain information about the hidden logical gate.

The present results provide evidence of a measurable, reproducible gate-dependent signal in this simulated H_L/S_L setting. They should not be interpreted as reliable practical single-shot gate identification. The inference result is modest, depends on the specific **surface-sim** implementation and noise model, and uses a Bernoulli Naive Bayes approximation rather than exact maximum-likelihood inference. The analysis is also restricted to two logical gates and has not been validated on hardware. The result is therefore best treated as a controlled baseline for the subsequent multi-gate SecureGate experiments.

Notebook navigation for review. The notebook is organized in the same sequence as this report: backend/API inspection $\rightarrow$ distance-5 layout and folding reference $\rightarrow$ independent H_L/S_L layout metadata $\rightarrow$ logical-circuit encoding $\rightarrow$ detector-coordinate verification $\rightarrow$ 10,000-shot fingerprint estimation $\rightarrow$ spatial/temporal fingerprint comparison $\rightarrow$ Bernoulli Naive Bayes inference $\rightarrow$ statistical testing and split robustness $\rightarrow$ label-permutation and fresh-shot controls $\rightarrow$ temporal ablation and detector-level model interpretation.

References.

1. S. Shukla, D. E. Browne, and S. Nishio, “Anticipating decoder side-channel attacks in fault-tolerant quantum computers,” arXiv:2607.12174 (2026). The paper introduces gate fingerprints as gate-dependent statistical structure in spacetime syndrome data and formulates gate inference from detector records.
2. The installed **surface-sim** logical-gate implementation records a folded transversal- H circuit reference at [arXiv:2406.17653](#) and an S -gate implementation reference at DOI [10.1088/1367-2630/17/8/083026](#); these references are retained here exactly as recorded by the backend source inspected in the notebook.